

Cultural Homophily in Tennis Doubles

Grand Slam Analysis 2018–2025

Prepared for supervisor review · May 2026 · Code: code/analysis/homophily_analysis.ipynb

Key Findings

- **Language proximity does not significantly predict match wins** after controlling for ranking quality. Ranking variables dominate match outcomes, as expected.
- **No robust pressure advantage under tiebreak stress:** language proximity is not statistically significant for tiebreak outcomes in the baseline specification.
- **Homophily shares are stable pre-2024** (~40% same-nationality, ~53% same-language from 2018–23), then drop sharply in 2024–25 — an artifact of expanded draw sizes (134 vs. ~62 matches per Grand Slam).
- **Ranking and career win rate dominate** match outcomes, as expected.

1. Data Overview

- **Matches:** 2,192 total | 2,128 Grand Slams | 64 Olympics
- **Years:** 2018, 2019, 2021, 2022, 2023, 2024, 2025 (2020 excluded — COVID)
- **Tournaments:** Australian Open, Roland Garros, Wimbledon, US Open, Olympics (2021, 2024)
- **Analysis sample:** 3,476 team-level observations (1,738 GS matches × 2 teams)

Player rankings and biographical profiles (nationality, language, coach) are sourced from ATP Tour snapshots merged to each match. Language variables derive from the CEPII Gravity dataset: *same language (official)* uses common official language; *language proximity* uses the broader ethno-linguistic criterion.

2. Pressure Outcomes

2.1 Tiebreak Observations

A **regular tiebreak** (7-point, played at 6-6 in sets 1 or 2) occurred in **44.0% of matches**. A **match tiebreak** (10-point super-tiebreak, replacing a full third set) was played in 30 matches, all at the Olympics or Wimbledon. For regression purposes both types are combined under *any tiebreak*.

Pressure Outcome	N	Share
Set-1 tiebreak (7-pt)	530	24.2%
Set-2 tiebreak (7-pt)	569	26.0%
Set-3 regular tiebreak (7-pt)	230	10.5%
Match tiebreak / super-tb (10-pt, set 3 ≥ 8)	30	1.4%
Any tiebreak (all types)	1,076	49.1%
Match went to 3 sets	903	41.2%
Comeback win (winner lost set 1)	395	18.0%

N = 2,192 total matches. Regular tiebreaks are most frequent at Wimbledon (52%) and least frequent at Roland Garros (35%).

2.2 By Tournament

Tournament	N Matches	Reg. Tiebreaks	TB Rate	Match Tiebreaks	Comeback Wins	Comeback Rate
Australian Open	580	262	45.2%	0	109	18.8%
Olympics	64	28	43.8%	20	13	20.3%
Roland Garros	581	201	34.6%	0	114	19.6%
US Open	375	165	44.0%	0	56	14.9%
Wimbledon	592	308	52.0%	10	103	17.4%

2.3 Comeback Wins

A comeback win is defined as the eventual match winner having lost the first set. This occurred in **395 matches (18.0%)** of the full sample, and in **43.7%** of three-set matches — meaning the team that won the first set subsequently lost the match roughly two-fifths of the time.

3. Cultural Homophily Across Olympic Cycles

The descriptive tables below show the share of pairs (averaged over both teams in each match) with each homophily characteristic, broken out separately for the Tokyo 2021 and Paris 2024 cycles. Grand Slam matches only — the Olympics tournament is excluded because same-nationality pairing is compulsory there.

3.1 Tokyo 2021 cycle

Table 2A. Tokyo 2021 cycle — Team composition across period

	Same Nationality (%)	Same Language (%)	Ling. Proximity (%)	N matches
Full Sample	37.3	50.2	50.7	2128.0
Pre-Tokyo (2018–19)	40.0	53.4	54.3	512.0
Tokyo Prep (2020–21)	41.4	54.3	54.3	187.0
Post-Tokyo (2021–22)	38.3	52.4	53.2	311.0

Tokyo panel includes 2020 as Tokyo Prep, excluding Wimbledon 2020 because Olympic doubles was not yet in the build-up path.

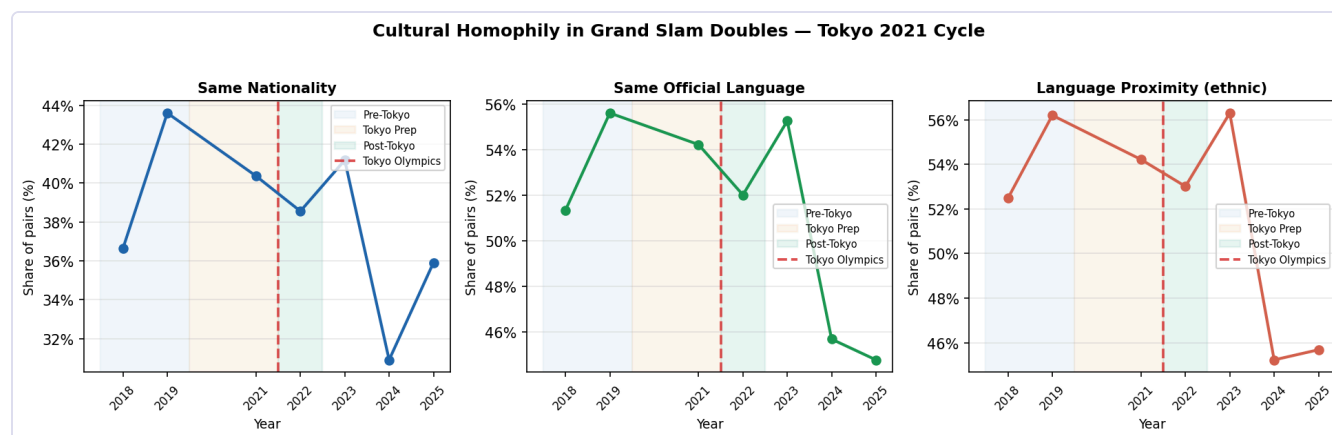


Figure 1. Tokyo 2021 cycle homophily trends. Shaded bands mark Pre-Tokyo, Tokyo Prep, and Post-Tokyo periods.

3.2 Paris 2024 cycle

Table 2B. Paris 2024 cycle — Team composition across period

	Same Nationality (%)	Same Language (%)	Ling. Proximity (%)	N matches
Full Sample	37.3	50.2	50.7	2128.0
Pre-Paris (2022)	38.6	52.0	53.0	249.0
Paris Prep (2023–24)	34.4	49.5	49.6	590.0
Post-Paris (2024–25)	35.8	45.0	45.7	528.0

Paris panel treats the 2023–24 Grand Slams as the Paris Prep window, with 2024 US Open and 2025 matches as Post-Paris.

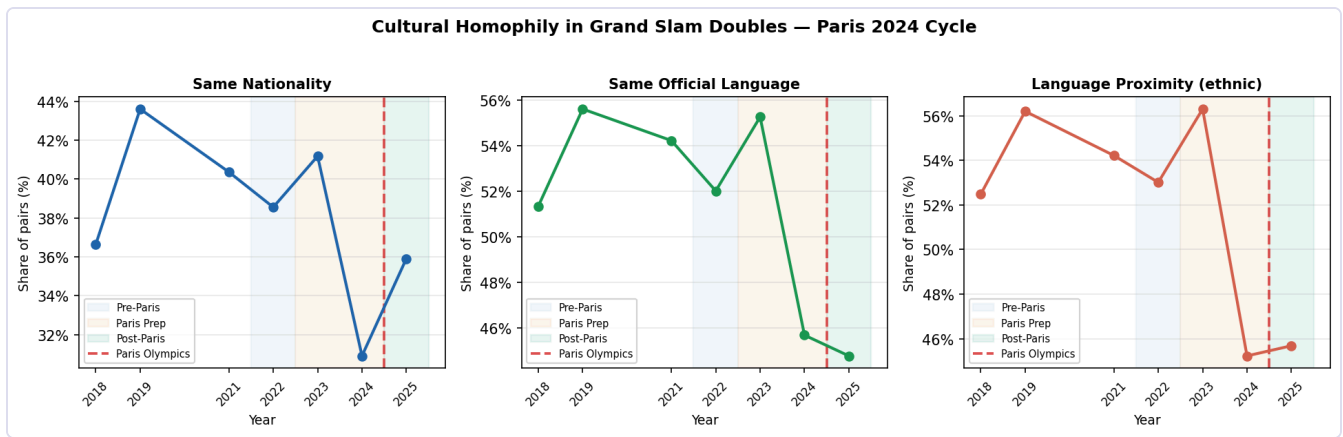


Figure 2. Paris 2024 cycle homophily trends. Shaded bands mark Pre-Paris, Paris Prep, and Post-Paris periods.

3.3 Winner vs. Loser Team Homophily

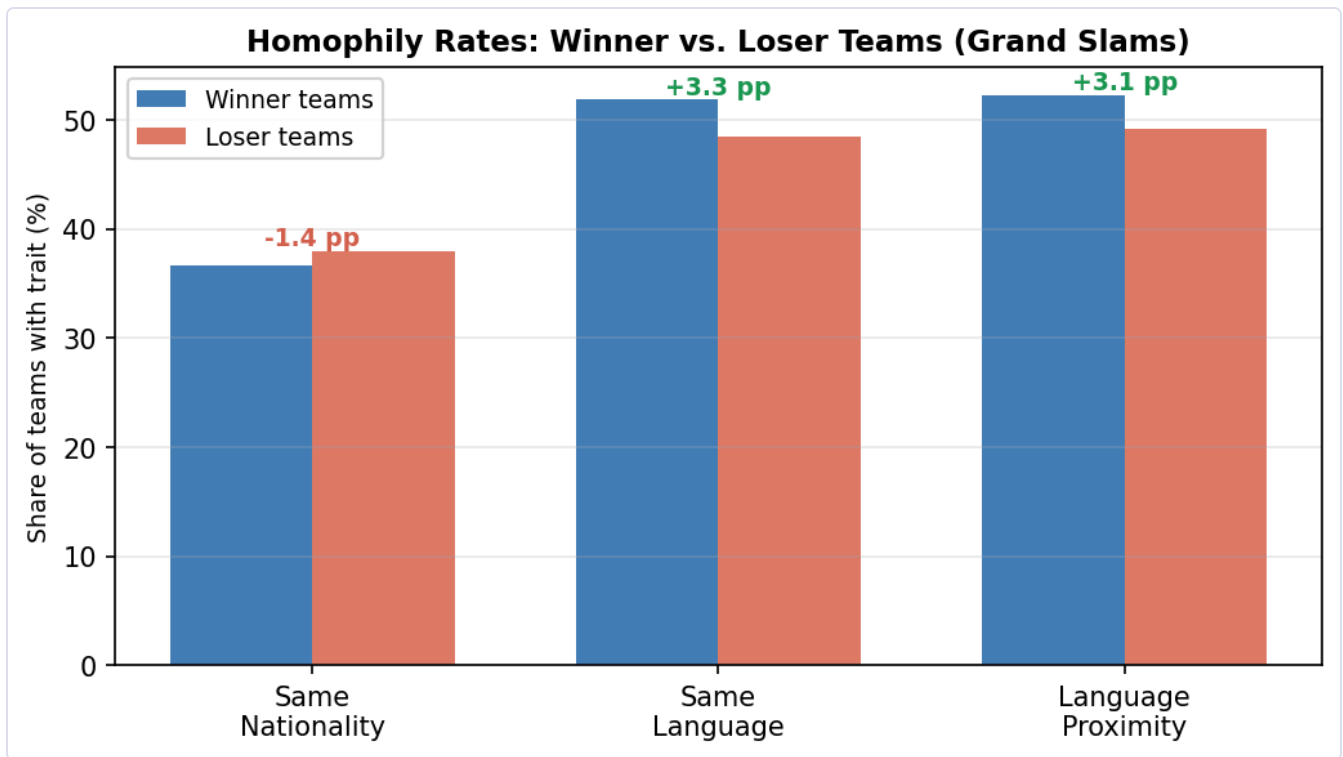


Figure 3. Percentage of teams with each homophily trait, split by match outcome. Annotation shows the winner-minus-loser gap.

Winner teams are more likely to share a language (+3.3 pp) and to be linguistically proximate (+3.1 pp) than loser teams. Same-nationality is marginally more common among losers (38.0% vs 36.6%).

4. Baseline Regressions

Specification: each match contributes two team-level observations (winner = 1, loser = 0). Regressors include language proximity (ethnic) and ranking controls (team mean ranking, opponent mean ranking, within-team rank gap, and a top-100 singles indicator). Fixed effects for tournament×year and round. Standard errors are clustered by match.

4.1 Match Win

	Logit	
Language proximity (ethnic)	+0.2756 ^{***}	(0.0722)
Team average doubles ranking	-0.0053 ^{***}	(0.0007)
Opponent average doubles ranking	+0.0031 ^{***}	(0.0004)
Teammate rank gap	+0.0014 ^{**}	(0.0006)
Top-100 singles player	+0.0427	(0.1429)
Observations		3,476
R ² / Pseudo-R ²		0.066

Table 3. Match Win. Outcome = 1 if team won. Tournament×year and round FE. SE clustered by match.

Significance: *** p < 0.01 ** p < 0.05 * p < 0.10 Standard errors in parentheses.

Language proximity is not statistically significant for match wins once rankings are controlled for.

4.2 Tiebreak Win (pressure outcome)

Sample restricted to matches where a regular tiebreak occurred in set 1 or 2 (N = 875 matches). Outcome: team won at least one tiebreak set.

	Logit	
Language proximity (ethnic)	+0.0847	(0.1096)
Team average doubles ranking	-0.0006	(0.0007)
Opponent average doubles ranking	+0.0011 ^{**}	(0.0005)
Teammate rank gap	-0.0007	(0.0006)
Top-100 singles player	-0.1877	(0.2148)
Observations		1,750
R ² / Pseudo-R ²		0.025

Table 4. Tiebreak Win — Logit (sample: matches with any tiebreak, 7p/10p). Outcome = 1 if team won any tiebreak. Tournament×year and round FE. SE clustered by match.

Significance: *** p < 0.01 ** p < 0.05 * p < 0.10 Standard errors in parentheses.

Language proximity is not statistically significant for tiebreak outcomes in the baseline specification.

4.3 Comeback Win (pressure outcome)

Sample: three-set Grand Slam matches *conditional on the team having lost set 1*. Outcome = 1 if the team wins the match (i.e., a comeback). Tournament×year fixed effects are included by encoding each tournament-year cell; cells with no within-cell variation in the outcome are dropped. Standard errors are clustered by match.

	Logit	
Language proximity (ethnic)	-0.3150 [*]	(0.1708)
Team average doubles ranking	-0.0034 ^{**}	(0.0014)
Opponent average doubles ranking	+0.0013	(0.0008)
Teammate rank gap	+0.0011	(0.0015)
Top-100 singles player	+0.4343	(0.3613)
Observations		735
R ² / Pseudo-R ²		0.095

Table 5. Comeback Win — Logit (3-set GS matches; conditional on losing set 1). Outcome = 1 if team wins given it lost set 1. Tournament×year FE (cells with no within-cell variation dropped). SE clustered by match.

Significance: *** p < 0.01 ** p < 0.05 * p < 0.10 Standard errors in parentheses.

Language proximity is not statistically significant for comebacks in this specification. Ranking controls have the expected signs: conditional on being a set down, stronger teams are more likely to complete the comeback.

5. Discussion and Next Steps

What the data say

- Language similarity has the most consistent positive sign across all three outcomes (match win, tiebreak, comeback), but never reaches conventional significance.
- Same nationality is actually slightly negative for match win in the descriptor (losers slightly more same-nationality), and the regression coefficient, while positive, is imprecisely estimated.
- Homophily shares do not rise in the run-up to the Olympics — if anything they fall in 2024–25, driven by draw expansion.

Limitations and extensions

- **Sample size:** ~1,700 Grand Slam matches. Power is limited for detecting small homophily effects (≤ 3 –5 pp).
- **Language collinearity:** same language and language proximity are highly correlated; consider a single index.
- **Draw expansion 2024–25:** the larger sample changes composition and may mask cycle-level patterns; restrict to comparable round ranges for a cleaner trend.
- **Conditioning on partnership formation:** selection — same-country pairs may self-select into tournaments — is not yet addressed.
- **Pair fixed effects:** a within-pair panel (same partners over time) would better isolate performance from selection.